

Hyper-parameters				Acc($\mathcal{M}$, $\mathcal{Q}$, $\mathcal{P}_{\text{irrelevant}}$)			Acc($\mathcal{M}$, $\mathcal{Q}$, $\mathcal{P}_{\text{inaccurate}}$)		
N	M	D	C	Base-9	Sym.(E)	Com.	Base-9	Sym.(E)	Com.
5	1	5	Y	57.7	38.7	55.3	53.3	39.7	51.0
5	1	5	N	54.7	32.7	53.7	47.0	32.3	55.7
5	2	2+3	Y	60.7	42.7	54.7	58.7	41.3	49.7
5	2	2+3	N	56.7	33.0	54.7	49.7	32.0	53.0
5	3	1+2+2	Y	60.7	38.7	53.3	58.0	43.3	49.0
5	3	1+2+2	N	56.0	33.3	55.7	48.7	32.0	52.3
5	5	1	Y	59.3	39.7	55.7	58.0	39.0	48.7
5	5	1	N	55.3	35.7	55.9	48.7	33.3	50.7

Table 9: Comparison of accuracy on medium-level tasks.

Hyper-parameters				#Tokens in step-3 (irr.)			#Tokens in step-3 (ina.)		
N	M	D	C	Base-9	Sym.(E)	Com.	Base-9	Sym.(E)	Com.
5	1	5	Y	1440	3162	788	1428	3170	798
5	1	5	N	1301	2685	660	1295	2732	667
5	2	2+3	Y	2175	4934	1269	2156	4989	1311
5	2	2+3	N	1864	4044	1005	1842	4087	1039
5	3	1+2+2	Y	2902	6704	1772	2878	6785	1821
5	3	1+2+2	N	2416	5360	1372	2393	5443	1420
5	5	1	Y	4368	10340	2764	4339	10514	2845
5	5	1	N	3535	8099	2088	3506	8303	2163

Table 10: Comparison of #tokens on medium-level tasks.

Model	Method	Acc($\mathcal{M}$, $\mathcal{Q}$, $\mathcal{P}_{\text{irrelevant}}$)			Acc($\mathcal{M}$, $\mathcal{Q}$, $\mathcal{P}_{\text{inaccurate}}$)		
		Base-9	Sym.(E)	Com.	Base-9	Sym.(E)	Com.
GPT-3.5-turbo	Base	30.3	25.1	42.3	10.1	26.1	33.4
	SC	36.6	28.3	45.0	17.3	30.7	44.7
	BT	34.2	22.7	28.3	18.4	22.7	11.0
	CC	44.3	32.7	43.6	31.7	33.0	40.8
	CD-CoT	60.7	42.7	54.7	58.7	41.3	49.7
Gemini-Pro	Base	72.3	38.9	53.2	21.2	36.7	33.5
	SC	80.3	43.3	60.0	32.3	45.0	42.7
	BT	82.4	29.3	37.8	26.7	28.7	33.3
	CC	67.5	37.3	50.2	43.6	35.0	45.6
	CD-CoT	92.7	49.3	57.7	76.7	53.3	55.7
LLaMA2-70B	Base	2.8	8.7	41.9	2.7	9.1	40.2
	SC	5.0	10.3	46.7	3.0	9.7	46.0
	BT	1.4	11.2	36.1	0.9	12.5	36.2
	CC	1.1	16.3	29.9	2.8	14.0	28.3
	CD-CoT	4.0	9.7	39.3	2.7	9.7	39.7
Mixtral-8x7B	Base	16.3	17.9	34.9	3.7	15.1	31.1
	SC	20.0	21.7	37.0	2.7	18.0	37.7
	BT	4.1	9.7	6.2	2.4	10.1	10.5
	CC	24.4	18.5	36.0	12.5	18.3	35.7
	CD-CoT	8.7	22.7	40.3	4.7	21.3	40.3

Table 11: Comparing methods with different LLMs.

Original Noisy Rationale	
In base-9, the digits are "012345678". <u>9 + 8 = 17</u> . We have 6 + 7 = 13 in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. 13 mod 9 = 4, so the digit is 4 and the carry is 1. We have 8 + 5 + 1 = 14 in base-10. <u>14 + 8 = 22</u> . 14 mod 9 = 5, so the digit is 5 and the carry is 1. A leading digit is 1. So the answer is 154. Answer: 154.	
Method	Denoised Rationale
SM	In base-9, the digits are "012345678". <u>9 + 8 = 17</u> . We5hav6r+ 7l=J13 in base-10. Since we're in base-9, ithFt exceeis the maCimum valur of 8 for a siLgle]digit. x0c3 mod 9 = , soWghe digix is 4 anj the carry is 1. We hOve 8 + 5i+ z2= 14 Vn baAe 10. <u>14 d 8 = 2p</u> . 14 mod 9 = 5, so Xhe diGit*iU 5 and the carry is 1Q A leading x0cigit is 1. So the answer is 154. Asswer: 154.
SD	In base-9, the digits are "012345678". <u>9 + 8 = 17</u> . We have 6 + 7 = 13 in base-10. Since we're in base-9, 13 exceeds the maximum value of 8 for a single digit. 13 mod 9 = 4, so the digit is 4 and the carry is 1. We have 8 + 5 + 1 = 14 in base 10. 14 mod 9 = 8. 14 div 9 = 1, so the digit is 8 and the carry is 1. A leading digit is 1. So the answer is 154. Answer: 154.
SP	In base-9, the digits range from 0 to 8. Let's add 86 and 57 step by step. First, we add the rightmost digits, which are 6 and 7. In base-10, 6 + 7 equals 13. However, since we are in base-9, the maximum value for a single digit is 8. Therefore, we need to find the remainder when 13 is divided by 9. The remainder is 4, so the digit in the units place is 4 and we have a carry of 1. Next, we add the next digits, which are 8 and 5, along with the carry of 1. In base-10, 8 + 5 + 1 equals 14. Again, we need to find the remainder when 14 is divided by 9. The remainder is 5, so the digit in the tens place is 5 and we have a carry of 1. Finally, we have a leading digit of 1. So the final answer in base-9 is 154. Answer: 154.
Ours	In base-9, the digits are "012345678". We have 6 + 7 = 13 in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. 13 mod 9 = 4, so the digit is 4 and the carry is 1. We have 8 + 5 + 1 = 14 in base 10. 14 mod 9 = 5, so the digit is 5 and the carry is 1. A leading digit is 1. So the answer is 154. Answer: 154.

Table 12: Comparison of rephrased rationales by different reasoning methods.

Ablation study with different LLMs. We examine the generalization of CD-CoT across different LLMs. As shown in Tab. 11, with comparably more powerful LLMs, *e.g.*, GPT-3.5-turbo and Gemini-Pro, CD-CoT demonstrates notable improvements in average accuracy. It respectively achieves increases of 23.4% and 21.6% in accuracy compared to base models and surpasses all the baselines.

Case Study. We illustrate the denoising effects of various robust methods using Math Base-9 as an example. The results in Tab. 12 indicate that the introduction of random characters by SM disrupts the logic of the rationale. SD fails to eliminate all the noise while recovering the input content, and SP alters the original rationale's reasoning process even when noise removal is successful. In contrast, CD-CoT significantly removes noisy thoughts and ensures format alignment with the original rationale. More denoising examples and an entire case study are in Appendix F.9 and G, respectively.

6 Conclusion

In this work, we investigate the under-explored problem of noisy rationales in LLMs. We introduce the NoRa dataset, which tests LLMs against irrelevant or inaccurate thoughts in question-answer scenarios. Our findings show LLMs' vulnerability to noisy rationales is inadequately mitigated by existing robust methods. We thereby design the CD-CoT method to enhance the robustness via contrastive denoising. The extension advocates for advancing LLMs by strategies, *e.g.*, external knowledge bases with a retrieval-augmented framework, robust inductive reasoning to extract rules from noisy examples, and multi-modal data integration to enhance the robustness of LLM reasoning.

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